

End-of-Summer Trend Analysis: Exponential Smoothing

Using 20 years of daily high temperature data for Atlanta (July through October) (file temps.txt), build and use an exponential smoothing model to help make a judgment of whether the unofficial end of summer has gotten later over the 20 years.

Methodology

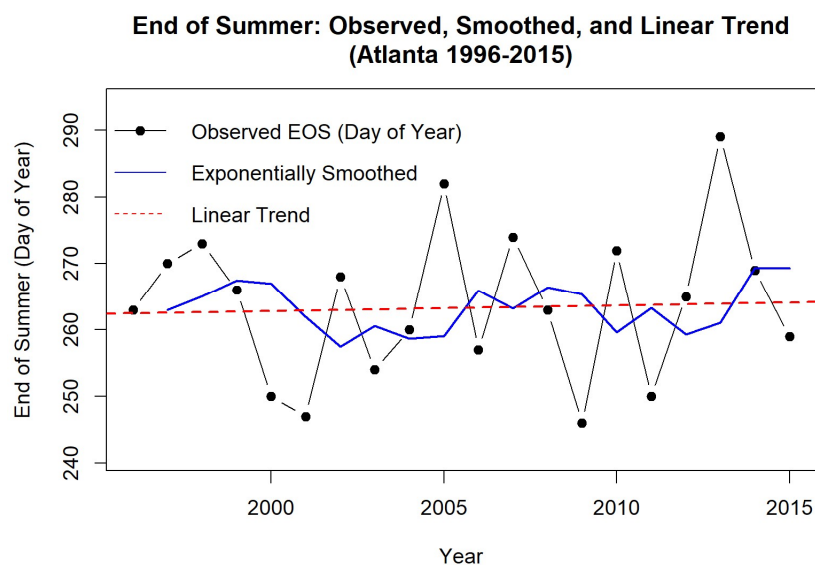
1. After reading in the temps.txt file using R, I confirmed 123 rows and 21 columns, with the "Day" column being the date, and every other column containing the average temperature for years 1996-2015.
2. The end-of-summer dates were reproduced using the same CUSUM logic described in the accompanying end-of-summer detection analysis, yielding one date per year for 1996-2015.
3. Each end-of-summer date was converted to a day-of-year value (e.g., September 19 = day 262) to produce a numeric time series of 20 annual values.
4. A time series object was created from these 20 day-of-year values with start=1996 and frequency=1, reflecting one observation per year.
5. R's HoltWinters() function was applied using simple exponential smoothing (beta=FALSE, gamma=FALSE), as there is no within-year seasonality in annual end-of-summer dates. When alpha was optimized automatically, HoltWinters set it to approximately 0.000066, effectively ignoring the data and outputting a near-constant value of ~263 for all years, making trend detection impossible. Alpha was therefore set manually to 0.3 to produce meaningful smoothing while still dampening year-to-year noise.
6. The smoothed values were confirmed to track the data meaningfully, ranging from ~258 to ~270 across the 20 years.
7. A linear regression model was fit to the smoothed day-of-year values to test whether the end of summer has been trending later over time.
8. CUSUM was also applied to the smoothed values using the same hyperparameter grid as in previous analyses (k and h as multiples of sigma) to provide a secondary check on the linear trend finding.

Hyperparameters	First_Crossing_Year	Max_CUSUM
$k = .5\sigma, h = 2\sigma$	2015	8.340
$k = 1\sigma, h = 2\sigma$	NA	4.631
$k = 1.5\sigma, h = 2\sigma$	NA	0.923

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$k = 1\sigma, h = 2.5\sigma$	NA	4.631
$k = 1.5\sigma, h = 2.5\sigma$	NA	0.923
$k = .5\sigma, h = 3\sigma$	NA	8.340
$k = 1\sigma, h = 3\sigma$	NA	4.631
$k = 1.5\sigma, h = 3\sigma$	NA	0.923

9. Reviewing the hyperparameter results, only one combination ($k=0.5\sigma, h=2\sigma$) produced a threshold crossing, and only in 2015 — the final year of the dataset. No consistency across combinations was observed, as crossings were effectively absent. Given the lack of meaningful CUSUM signal, no hyperparameter combination stands out as clearly preferable. For completeness, $k=1\sigma$ and $h=2.5\sigma$ were selected as moderate values, consistent with the approach used in the previous analysis with this data.
10. A final CUSUM model was fit using $k=(1\sigma)$ and $h=(2.5\sigma)$. A linear model was fit to the smoothed end-of-summer day-of-year values directly, as the CUSUM signal was too weak to drive meaningful trend analysis on its own.
11. The results were plotted showing the observed end-of-summer dates, the exponentially smoothed values, and the linear regression trend line.

Results



For the linear regression model:

- Residual standard error: 3.782 on 17 degrees of freedom
- Multiple R-squared: 0.01764, Adjusted R-squared: -0.04015
- F-statistic: 0.3052 on 1 and 17 DF, p-value: 0.5878
- Linear trend slope of smoothed end-of-summer day-of-year values: 0.088

Using simple exponential smoothing on the end-of-summer dates, the analysis does not show that the unofficial end of summer has gotten later in Atlanta between 1996 and 2015.

Discussion of Results

The CUSUM model produced no sustained threshold crossing across the smoothed end-of-summer dates, with only the most sensitive combination ($k=0.5\sigma$, $h=2\sigma$) triggering a crossing, and only in 2015, the final year of the dataset. The linear regression model results confirm this:

- The slope of the linear regression line is 0.088, meaning the smoothed end-of-summer date increases by less than one tenth of a day per year — effectively flat.
- The linear model resulted in $p=0.588$, meaning there is a 58.8% chance of seeing a slope this size from random year-to-year variability alone. This is far above any conventional significance threshold.
- The $R^2=0.018$ indicates the linear trend explains less than 2% of the total year-to-year variability in smoothed end-of-summer dates.

Taken together, the exponential smoothing and linear trend analysis do not support the conclusion that the unofficial end of summer has gotten later in Atlanta over the 1996-2015 period. The end-of-summer dates are variable from year to year but show no meaningful directional trend.

References

The methodology and assessment I used was informed by:

- [Medium, \$R^2\$, Adjusted \$R^2\$, and p-values explained \(by Gaypen Guinse X\)](#)